INTELLIGENT DOCUMENT AUDITING

Confidence-Aware OCR & Multimodal Anomaly Detection

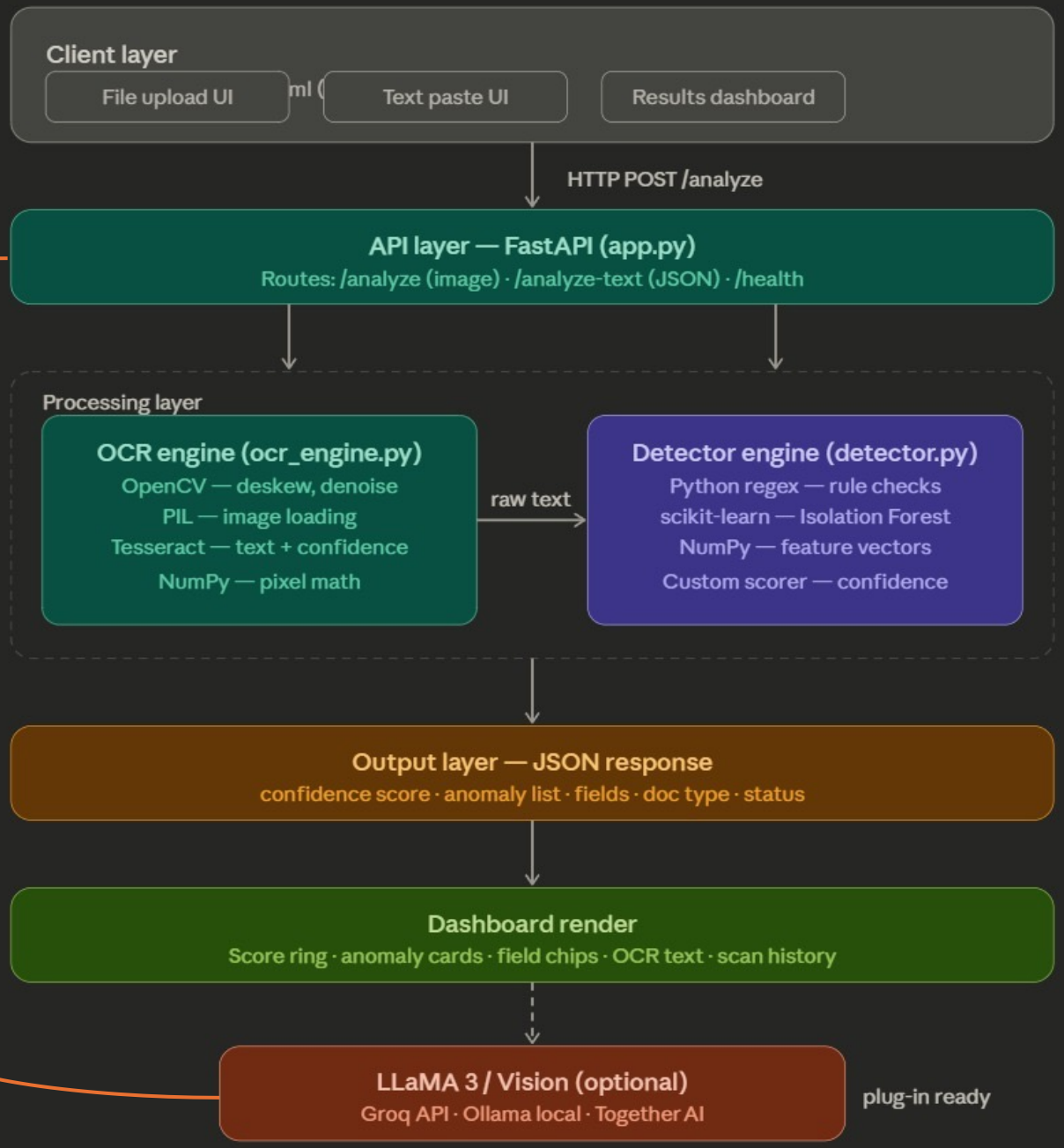
OCR and its Problems

- Optical Character Recognition (OCR) is the foundational technology used to digitize and extract raw text from physical or scanned documents
- While unmatched in speed and efficiency, standard OCR lacks visual and contextual awareness—which is exactly where our architectural override steps in
- Drawbacks compared to Manual Drafting :
  - A character with excessive curves or strike through might be inferred in a incorrect way
  - The words with similar characters might become indistinguishable ,changing the context of the document("Price" might be inferred as "Pride")
  - Important Verification Artifacts (signatures and stamps) might just be discarded or inferred as text
  - Document might have variables or values which do not support real-world scenario such as 32nd day of any month

Architecture Diagram

FastAPI has been utilized to support enterprise-level document analysis (around 1,500 pages per min)

It is an optional plugin used for cross-verifying and reducing the error percentage



OCR package

Primary Goal: Converts image pixels into raw text strings and numeric character-level confidence scores.

Modules Used:

PyTesseract: Standard fast engine to scan text and output character-level confidence percentages.

Llama 3.2 via Ollama: Local visual-language model to read document context and output structured JSON.

OpenCV (Image Morphology): Computer vision to isolate continuous pixels (bounding boxes) for signatures.

Output: A dictionary of raw text and crop files.

Detector package

Primary Role: Scans the raw text to catch anomalies, semantic errors, and fraud indicators and also arranges the text in correct format

Modules Used:

Scikit-Learn (IsolationForest): Unsupervised ML used to score statistical outliers without needing labelled data.

Python regex (RegEx): Hard-coded string pattern parsers to validate standard date schemas and serial codes.

Standard Python Math Operators: Floating-point precision scripts to check if Subtotal + Tax=Grand Total.

Output: The final formatted text

Drawbacks of OCR Solved !!

- Confidence Scoring Protocol (CSP) : Collection of the Confidence Score for each character scanned from the Tesseract and only validate the character if and only if it satisfies the Confidence Threshold Value

- Context - Evaluating Model : Validating if the words scanned in a sentence makes sense by using N – Gram Analysis and also predict flagged words by CSP

- Non – Textual Artifact Segmentation (Experimental) : Isolating signatures and stamps which add proof to the document authenticity using OpenCV

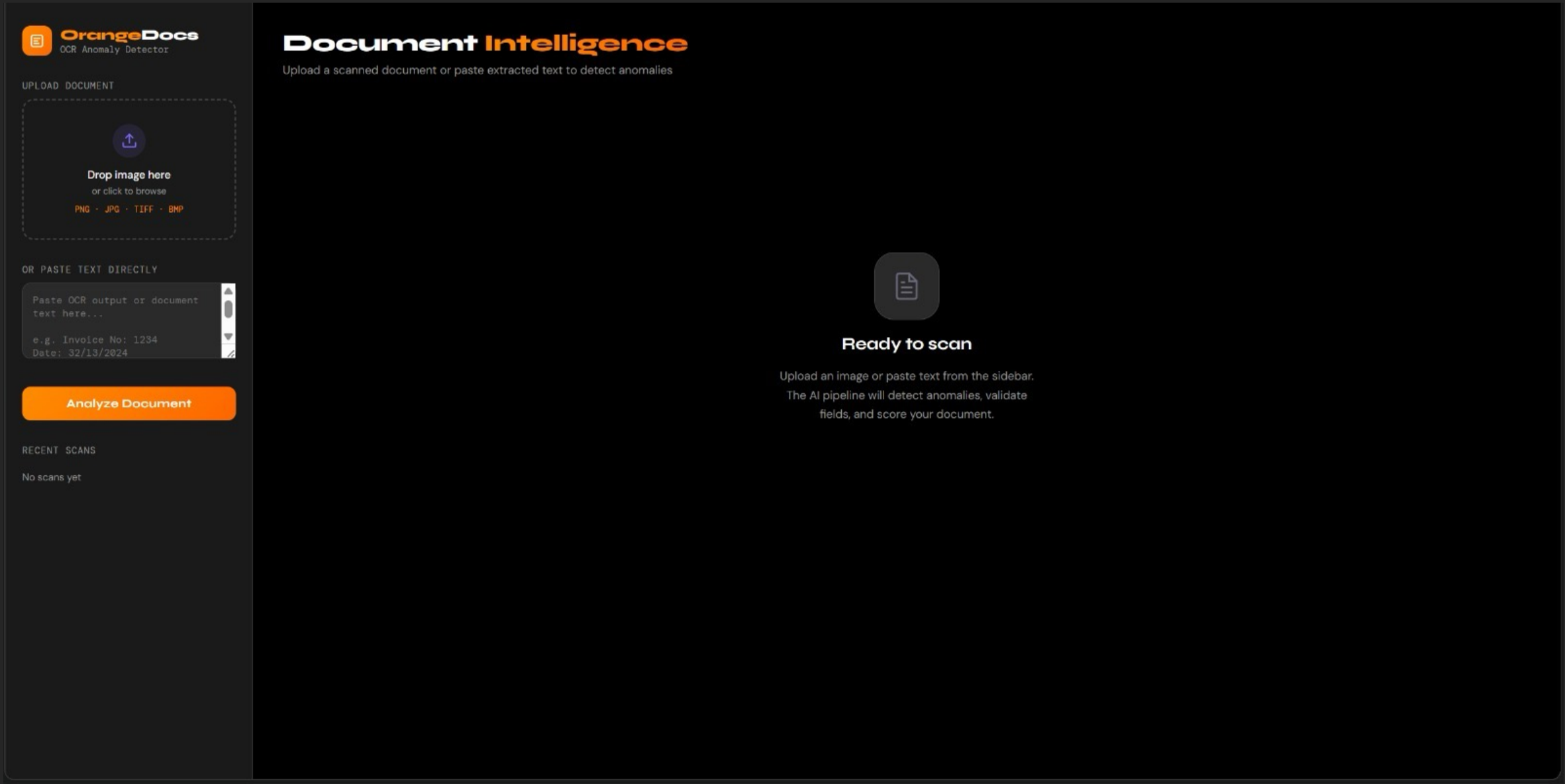
- Logical & Mathematical Constraints : Ensuring accounting and logical laws add up by using Specialized Detector Engine

Managing Real – World Constraints

- Computing and Hardware Constraints : Not all have advanced GPU's and Handling Capability which reduces the OCR to Enterprise – Only
  - Solution : We use local , quantized version of LLaMa 3.2 Vision via Ollama and Tesseract to keep Computing costs low

- Data Privacy: As we are dealing with confidential files with sensitive data , it cannot be cloud – based
  - Solution : We run Tesseract, Python scikit-learn, and local models on-premise, enterprise financial data never leaves your company server

Prototype of Client Interface



Academic & Industry Citations

"Isolation Forest" by Liu, F., Ting, K. M., & Zhou, Z. H. (2008) IEEE International Conference on Data Mining (ICDM). *(The foundational paper for detector.py outlier math).*

"Confidence-Aware Document Processing: Fusing Visual Uncertainty with Language Embeddings." by Hemmer, P., et al. (2024)
(The concept behind setting Tesseract threshold cut-offs).

"Hard Constraints vs. LLM Hallucinations in IDP Pipelines" by Kili Technology Whitepapers (2025-2026)
(The concept behind enforcing hard RegEx math over blindly trusting visual AI output).

THANK YOU !